Intermittent 30-second stalls on customer file uploads

INC-5455 · post-incident review, draft 3 · M. Dubois, Cloud Storage Platform · 2026-03-30

Jira issue	INC-5455 · investigation STO-1180, actions STO-1181 to STO-1185
Severity	SEV-3 — degraded, intermittent, self-recovering
Environment	Production — upload-svc v3.11.0, object store cluster obj-eu-2, F5 load balancers
First observed	2026-03-02 (retrospectively; first customer report 2026-03-09)
Last observed	2026-03-27 08:41 CET
Frequency	14 episodes over 25 days, each 28–34 seconds, no pattern by hour or weekday
Status	**Open** — monitoring in place, cause not established

1. Description

Between 2 and 27 March 2026, customer file uploads through upload-svc stalled intermittently. During an episode, an upload that had been streaming normally stopped transferring for 28 to 34 seconds and then resumed at full speed and completed successfully. The client showed a frozen progress bar. Uploads started during an episode were also affected; downloads, listing and deletion were not.

Fourteen episodes were confirmed from server-side traces. Each affected every upload in flight on one of the six upload-svc nodes, but only ever one node at a time, and never the same node twice in succession. 9,240 uploads were delayed in total. Eleven uploads — all from clients with a 30-second timeout — failed outright and had to be retried by the customer. No file was corrupted or partially stored: the object store either had the complete object or nothing.

2. Impact

- 9,240 uploads delayed by roughly 30 seconds across 14 episodes.
- 11 uploads failed and were retried manually by customers.
- 6 support tickets, 2 escalated by named enterprise accounts.
- No data loss and no corruption. Upload SLO (99.5% under 60s for a 10 MB file) breached in March: 99.31%.

Episode duration in seconds, by day in March 2026

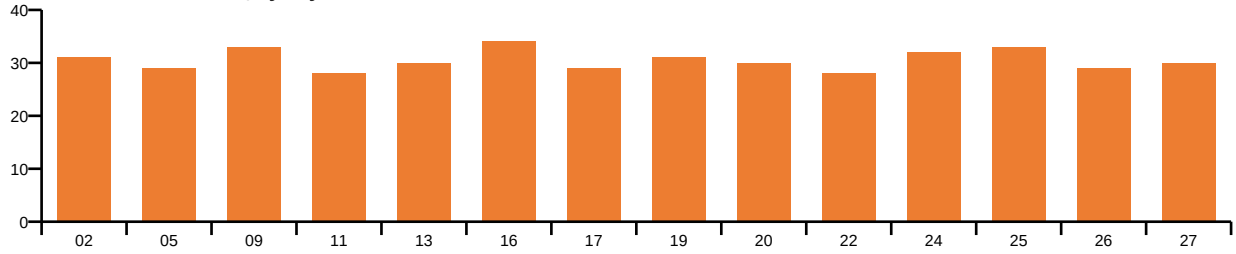

Figure 1 — duration is remarkably consistent at about 30 seconds, which is itself the strongest clue and is discussed under hypothesis B.

3. Timeline (CET)

When	What happened
2026-03-02 14:22	First episode, visible only in traces. No alert; the upload latency alert averages over 5 minutes and a 30-second stall on one node does not move the average enough to fire.
2026-03-09 10:05	First customer report. Treated as an isolated client network problem and closed.
2026-03-13 09:30	Third report. M. Dubois correlates the reports with trace data and finds nine prior episodes. INC-5455 raised.
2026-03-16 11:00	Investigation begins. Continuous packet capture enabled on all six upload nodes.
2026-03-17 15:48	Episode captured in full for the first time. The capture shows the node stops reading from its sockets; TCP windows go to zero; no error is logged.
2026-03-19	GC logs, thread dumps and kernel scheduler traces reviewed for the 17 March episode. No stop-the-world pause longer than 180 ms. Hypothesis A weakened.
2026-03-20 02:00	Object store vendor engaged with the captures. Vendor confirms no server-side throttling and no errors on their side during the episode windows.
2026-03-23	F5 load balancer firmware and connection-reuse settings reviewed. A 30-second idle timeout is found on the internal pool and is the basis of hypothesis B.

When	What happened
2026-03-25 16:00	Review with SRE, networking and the vendor. No hypothesis can be confirmed with the evidence held.
2026-03-26 09:00	Connection pool recycling changed as a mitigation (STO-1181), deployed to two of six nodes as a controlled experiment.
2026-03-27 08:41	Fourteenth and last episode — on one of the four unchanged nodes.
2026-03-30	Review held. Document issued with the cause unestablished.

4. Root cause

> **The root cause of this incident has not been established.** Three hypotheses were investigated. None could be confirmed or ruled out with the evidence available, and no episode has occurred since 27 March, so there is nothing left to observe. This document is issued with the cause open rather than held indefinitely.

	Hypothesis	Evidence for	Evidence against	Verdict
A	Garbage collection pause in upload-svc	Stalls affect one JVM at a time and end without intervention, which fits a stop-the-world pause.	GC logs for the captured episode show no pause above 180 ms. A 30-second pause would be logged.	Unlikely, not excluded
B	F5 idle timeout on the internal connection pool	The pool idle timeout is exactly 30 seconds, matching the observed duration closely.	The stall affects connections that are actively transferring, which should never be idle. The vendor cannot explain this.	Plausible, unproven
C	Silent throttling or a failover in the object store	Only one node is affected at a time, consistent with per-client throttling.	The vendor's own telemetry shows no throttling and no errors during any episode window. We cannot audit this independently.	Unproven, cannot verify

What is known with confidence: during an episode the affected node stops reading from its client sockets, no error is logged by any component, and the condition clears itself after approximately 30 seconds. What is not known: why it stops reading. The remaining actions are therefore weighted towards evidence collection rather than repair.

5. Action plan

Jira	Action	Type	Owner	Due	Status
STO-1181	Recycle pooled connections every 20 seconds so that no connection can reach the 30-second idle timeout. Mitigates hypothesis B without confirming it.	Mitigate	M. Dubois	2026-03-26	Done (2 of 6 nodes)
STO-1182	Roll the connection recycling change to the remaining four nodes and hold the configuration for 60 days.	Mitigate	M. Dubois	2026-04-03	Open
STO-1183	Alert on any single upload node whose socket read throughput falls to zero for more than 5 seconds, so that the next episode pages immediately rather than being found in traces.	Detect	A. Novak	2026-04-10	Open
STO-1184	Keep continuous packet capture and JFR recording enabled on all upload nodes until 2026-06-30, with a 7-day retention.	Investigate	M. Dubois	2026-04-03	Open
STO-1185	Request per-connection server-side telemetry from the object store vendor under the support contract, so hypothesis C can be tested rather than taken on trust.	Investigate	R. Bianchi	2026-04-24	Open

Review decision (2026-03-30). INC-5455 stays open at SEV-3 with no further investigation resourced. If a fifteenth episode occurs, STO-1183 will page and the incident returns to active investigation with the instrumentation from STO-1184 already in place. If none occurs by 30 June 2026 the incident will be closed with the cause recorded as undetermined.